

1 Basic convergence tests

Question 1. Match the series on the left with the appropriate convergence test and conclusion on the right, evaluating the series if asked.

$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$	Diverges by the divergence test
$\sum_{n=1}^{\infty} \frac{1}{\sqrt{1+n}}$	It's a convergent geometric series that converges to _____
$\sum_{n=2}^{\infty} \frac{1}{n \ln n}$	It's a telescoping series that converges to _____
$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$	Converges by the p -series Test
$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$	Diverges by the p -series Test
$\sum_{n=1}^{\infty} \frac{3^n}{4^n}$	Converges by the integral test
$\sum_{n=1}^{\infty} \frac{3}{n(n+2)}$	Diverges by the integral test

Question 2. The following boxed claim and proof are incorrect. Identify the mistake(s).

Claim: $\sum_{n=1}^{\infty} n^3 e^{-n} \leq 6$

Proof: The integral test tells us that

$$\int_1^{\infty} x^3 e^{-x} dx \leq \sum_{n=1}^{\infty} n^3 e^{-n} \leq \int_0^{\infty} x^3 e^{-x} dx.$$

We evaluate the right integral to find that

$$\int_0^{\infty} x^3 e^{-x} dx = [(-x^3 - 3x^2 - 6x - 6)e^{-x}]_0^{\infty} = 6.$$

Hence, the inequality holds.

Challenge: correctly prove that $\sum_{n=1}^{\infty} n^3 e^{-n}$ converges and find an actual upper bound. (Check your answer with the fact that $\sum_{n=1}^{\infty} n^3 e^{-n} \approx 6.0065$.)

2 Estimating errors

Let $a_n \geq 0$ be a nonnegative sequence. By estimating the infinite series $\sum_{n=1}^{\infty} a_n$ with a partial sum $\sum_{n=1}^N a_n$ to a precision of ϵ , we mean finding an integer N such that

$$\sum_{n=1}^{\infty} a_n - \sum_{n=1}^N a_n = \sum_{n=N+1}^{\infty} a_n < \epsilon.$$

This can be done by comparing the series to an integral. If f is a decreasing function such that $f(n) = a_n$, then integral test tells us that

$$\int_{N+1}^{\infty} f(x) dx \leq \sum_{n=N+1}^{\infty} a_n \leq \int_N^{\infty} f(x) dx,$$

so the task becomes finding an integer N such that

$$\sum_{n=N+1}^{\infty} a_n \leq \int_N^{\infty} f(x) dx < \epsilon.$$

Question 3 (Estimating errors). Estimate a value of N such that $\sum_{n=1}^N a_n$ approximates the infinite series $\sum_{n=1}^{\infty} a_n$ to within a precision of 0.03.

a) $a_n = \frac{1}{n^2}$

$$\text{b) } a_n = \frac{1}{(n+1)(\ln(n+1))^2}$$

c) $a_n = \frac{1}{n(n+1)}$ (Challenge: use another method besides comparison with an integral)

3 Comparison tests

Question 4. The following boxed claim and proof are incorrect. Identify the mistake(s).

Claim: $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n}$ diverges.

Proof: Since $\sin(1/n) \leq 1$ for every positive integer n , we can apply direct comparison test with the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since the harmonic series diverges, the original series must also diverge.

Challenge: show that $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n}$ converges.

Question 5. Match the series with the appropriate conclusion and state the series used to compare.

$$\sum_{n=1}^{\infty} \frac{n}{3^n}$$

Converges because it grows at the same rate as the convergent series:

$$\sum_{n=1}^{\infty} \frac{n+2}{n^3-2n+2}$$

Diverges because it grows at the same rate as the divergent series:

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

Converges because it grows slower than the convergent series:

$$\sum_{n=1}^{\infty} \frac{1}{1 + \ln n}$$

Diverges because it grows faster than the divergent series: